

K-S Implication-Only Exercise Bank Up to Permutation

Executive summary

I treated your request as an internal proof-search / type-inhabitation problem for the implication-only K,S Hilbert system and ran a breadth-first enumeration by exact proof-tree MP-count. The key headline result is:

- **Permutation first appears at MP-depth 9:**

- $P: (A \rightarrow B \rightarrow C) \rightarrow B \rightarrow A \rightarrow C$

The raw search grows very fast. After removing all formulas that are already a single axiom instantiation, there are:

- **8,531** nontrivial schemata strictly below P across depths 1 through 8
- **17,371** more at depth 9 alone

Because printing the whole raw bank would be unusable, I report an **exhaustive small-size frontier**:

- **all non-axiom theorem schemata of size ≤ 11 that are strictly below P**
- this gives **182 exercises**, fully ordered by:
 - minimal MP-depth
 - minimal formula size
 - lexicographic order on canonical variable names

That is enough to give you a serious, breadth-first training bank while still staying readable. I mark the precise threshold where P appears, then give a compact post-threshold starter set.

The point where repetition begins is **right after size 11**. Below P , the counts by size are:

- size 3 : 1
- size 5 : 1
- size 7 : 13
- size 9 : 46
- size 11 : 121

Then the next size layer jumps to:

- size 13 : 211
- size 15 : 348
- size 17 : 452
- size 19 : 543

• size 21 : 574

That is the natural spot to stop the exhaustive printed list and switch to threshold / post-threshold material.

Methodology

I modeled the system exactly as the implication-only Hilbert calculus with axiom schemata

- K: $A \rightarrow B \rightarrow A$
- S: $(A \rightarrow B \rightarrow C) \rightarrow (A \rightarrow B) \rightarrow A \rightarrow C$

and modus ponens as the sole inference rule. I then enumerated **closed SK applicative terms** breadth-first by number of application nodes, computed each term's principal implicational type by freshening and unification, canonicalized variable names, and recorded the **first depth** at which each schema appears. Under propositions-as-types, the types of K and S correspond to Hilbert axiom schemata, and term application corresponds to modus ponens, so **application-count is exactly proof-tree MP-count** for this search. That is the reason a combinator-term BFS gives the right “genuine proof difficulty” ordering here. Standard Hilbert presentations also stress that implication-only systems with only modus ponens are simple to define but awkward for backward proof search, which is exactly why an explicit BFS criterion matters. ¹

Canonicalization and filtering were done as follows.

- **Right association** throughout: $A \rightarrow B \rightarrow C$ means $A \rightarrow (B \rightarrow C)$.
- **Formula size** means syntax-tree node count: each variable occurrence counts 1, each $\rightarrow$ node counts 1.
- **Single-step filter**: I removed every schema that is an instance of K or S, not just the general forms themselves.
- **Duplicate control**: formulas were canonicalized by renaming variables in order of first occurrence, so alpha-renamings collapse to one schema.
- **Printed cutoff**: I printed the exhaustive bank only through size 11. Beyond that, the frontier becomes dominated by size-inflated variants of earlier patterns, and the count jumps sharply, so the marginal pedagogical value per extra page drops fast.

Counts and threshold

Raw search counts by MP-depth

These are the counts of **all** non-axiom theorem schemata found at each exact MP-depth, with no size cutoff.

MP-depth	Raw non-axiom schemata
1	4
2	10
3	29
4	74

MP-depth	Raw non-axiom schemata
5	207
6	632
7	1,883
8	5,692
9	17,371

So the strict-below- P total is:

$$\bullet 4 + 10 + 29 + 74 + 207 + 632 + 1,883 + 5,692 = 8,531$$

Printed bank counts by MP-depth

This is the **reported** bank: all non-axiom schemata of size ≤ 11 .

MP-depth	Printed bank count
1	2
2	6
3	12
4	7
5	15
6	33
7	49
8	58
9	85

Printed below-threshold counts by size

This table is just the bank **strictly below** P that is fully printed in the next section.

Formula size	Count below P
3	1
5	1
7	13
9	46

Formula size	Count below P
11	121

Total printed below P: **182**

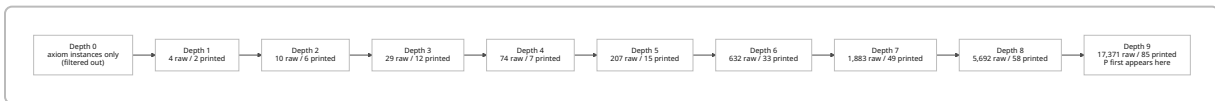
Cutoff and repetition point

The printed list is exhaustive through **size 11**. I stopped the exhaustive print there because the next pre-threshold layers already expand to:

Formula size	Raw count below P
13	211
15	348
17	452
19	543
21	574

That is the point where the search starts yielding many lengthened variants of the same basic proof motifs. This is the explicit “repetition begins here” marker.

BFS layering sketch



First twenty exercises

These are the first twenty items of the printed bank under the required ordering.

Index	Formula	MP-depth	Formula size
1	$(A \rightarrow B) \rightarrow A \rightarrow A$	1	7
2	$A \rightarrow B \rightarrow C \rightarrow B$	1	7
3	$A \rightarrow A$	2	3
4	$(A \rightarrow B) \rightarrow A \rightarrow C \rightarrow B$	2	9
5	$A \rightarrow (B \rightarrow C) \rightarrow B \rightarrow B$	2	9
6	$A \rightarrow B \rightarrow C \rightarrow D \rightarrow C$	2	9

Index	Formula	MP-depth	Formula size
7	$((A \rightarrow B) \rightarrow A) \rightarrow (A \rightarrow B) \rightarrow A$	2	11
8	$(A \rightarrow B \rightarrow C) \rightarrow A \rightarrow B \rightarrow C$	2	11
9	$A \rightarrow B \rightarrow B$	3	5
10	$(A \rightarrow B) \rightarrow A \rightarrow B$	3	7
11	$A \rightarrow B \rightarrow C \rightarrow A$	3	7
12	$(A \rightarrow A \rightarrow B) \rightarrow A \rightarrow B$	3	9
13	$((A \rightarrow B) \rightarrow A) \rightarrow (A \rightarrow B) \rightarrow B$	3	11
14	$(A \rightarrow (B \rightarrow A) \rightarrow C) \rightarrow A \rightarrow C$	3	11
15	$(A \rightarrow B \rightarrow C) \rightarrow A \rightarrow B \rightarrow B$	3	11
16	$(A \rightarrow B) \rightarrow (C \rightarrow A) \rightarrow C \rightarrow B$	3	11
17	$(A \rightarrow B) \rightarrow A \rightarrow C \rightarrow D \rightarrow C$	3	11
18	$A \rightarrow (B \rightarrow C) \rightarrow B \rightarrow D \rightarrow C$	3	11
19	$A \rightarrow B \rightarrow (C \rightarrow D) \rightarrow C \rightarrow C$	3	11
20	$A \rightarrow B \rightarrow C \rightarrow D \rightarrow E \rightarrow D$	3	11

Exercise bank below permutation

The bank below is **exhaustive for all non-axiom theorem schemata of size ≤ 11 that are strictly easier than $\mathbf{P}$** , ordered by minimal MP-depth, then size, then lexicographically.

Depth 1 — 2 exercises

- $(A \rightarrow B) \rightarrow A \rightarrow A$
- $A \rightarrow B \rightarrow C \rightarrow B$

Depth 2 — 6 exercises

- $A \rightarrow A$
- $(A \rightarrow B) \rightarrow A \rightarrow C \rightarrow B$
- $A \rightarrow (B \rightarrow C) \rightarrow B \rightarrow B$
- $A \rightarrow B \rightarrow C \rightarrow D \rightarrow C$
- $((A \rightarrow B) \rightarrow A) \rightarrow (A \rightarrow B) \rightarrow A$
- $(A \rightarrow B \rightarrow C) \rightarrow A \rightarrow B \rightarrow C$

Depth 3 — 12 exercises

1. $A \rightarrow B \rightarrow B$
2. $(A \rightarrow B) \rightarrow A \rightarrow B$
3. $A \rightarrow B \rightarrow C \rightarrow A$
4. $(A \rightarrow A \rightarrow B) \rightarrow A \rightarrow B$
5. $((A \rightarrow B) \rightarrow A) \rightarrow (A \rightarrow B) \rightarrow B$
6. $(A \rightarrow (B \rightarrow A) \rightarrow C) \rightarrow A \rightarrow C$
7. $(A \rightarrow B \rightarrow C) \rightarrow A \rightarrow B \rightarrow B$
8. $(A \rightarrow B) \rightarrow (C \rightarrow A) \rightarrow C \rightarrow B$
9. $(A \rightarrow B) \rightarrow A \rightarrow C \rightarrow D \rightarrow C$
10. $A \rightarrow (B \rightarrow C) \rightarrow B \rightarrow D \rightarrow C$
11. $A \rightarrow B \rightarrow (C \rightarrow D) \rightarrow C \rightarrow C$
12. $A \rightarrow B \rightarrow C \rightarrow D \rightarrow E \rightarrow D$

Depth 4 — 7 exercises

1. $A \rightarrow B \rightarrow C \rightarrow C$
2. $(A \rightarrow B) \rightarrow A \rightarrow C \rightarrow A$
3. $(A \rightarrow B) \rightarrow C \rightarrow A \rightarrow A$
4. $A \rightarrow (B \rightarrow C) \rightarrow B \rightarrow C$
5. $A \rightarrow B \rightarrow C \rightarrow D \rightarrow B$
6. $(A \rightarrow B \rightarrow C) \rightarrow D \rightarrow E \rightarrow D$
7. $A \rightarrow (B \rightarrow B \rightarrow C) \rightarrow B \rightarrow C$

Depth 5 — 15 exercises

1. $((A \rightarrow B \rightarrow A) \rightarrow A) \rightarrow A$
2. $((A \rightarrow B \rightarrow A) \rightarrow C) \rightarrow C$
3. $(A \rightarrow A \rightarrow A) \rightarrow A \rightarrow A$
4. $(A \rightarrow A \rightarrow B) \rightarrow A \rightarrow A$
5. $(A \rightarrow B) \rightarrow A \rightarrow C \rightarrow C$
6. $(A \rightarrow B) \rightarrow C \rightarrow D \rightarrow C$
7. $A \rightarrow B \rightarrow C \rightarrow D \rightarrow D$
8. $((A \rightarrow B \rightarrow A) \rightarrow C) \rightarrow D \rightarrow C$
9. $(A \rightarrow A \rightarrow B) \rightarrow A \rightarrow A \rightarrow B$
10. $(A \rightarrow B) \rightarrow A \rightarrow C \rightarrow D \rightarrow B$
11. $(A \rightarrow B) \rightarrow C \rightarrow A \rightarrow D \rightarrow B$
12. $A \rightarrow (B \rightarrow C) \rightarrow B \rightarrow D \rightarrow B$
13. $A \rightarrow (B \rightarrow C) \rightarrow D \rightarrow B \rightarrow B$
14. $A \rightarrow B \rightarrow (C \rightarrow D) \rightarrow C \rightarrow D$
15. $A \rightarrow B \rightarrow C \rightarrow D \rightarrow E \rightarrow C$

Depth 6 — 33 exercises

1. $(A \rightarrow A) \rightarrow A \rightarrow A$
2. $A \rightarrow (A \rightarrow B) \rightarrow B$

3. $((A \rightarrow A) \rightarrow B) \rightarrow A \rightarrow A$
4. $((A \rightarrow B) \rightarrow C) \rightarrow B \rightarrow C$
5. $(A \rightarrow B \rightarrow C) \rightarrow A \rightarrow A$
6. $(A \rightarrow B \rightarrow C) \rightarrow D \rightarrow D$
7. $A \rightarrow (A \rightarrow B) \rightarrow C \rightarrow B$
8. $A \rightarrow (B \rightarrow C) \rightarrow B \rightarrow A$
9. $A \rightarrow B \rightarrow (B \rightarrow C) \rightarrow B$
10. $A \rightarrow B \rightarrow C \rightarrow D \rightarrow A$
11. $((A \rightarrow B) \rightarrow A \rightarrow A) \rightarrow C \rightarrow C$
12. $((A \rightarrow B \rightarrow A) \rightarrow A) \rightarrow B \rightarrow A$
13. $((A \rightarrow B \rightarrow C \rightarrow B) \rightarrow A) \rightarrow A$
14. $((A \rightarrow B \rightarrow C \rightarrow B) \rightarrow D) \rightarrow D$
15. $((A \rightarrow B) \rightarrow A) \rightarrow C \rightarrow D \rightarrow C$
16. $((A \rightarrow B) \rightarrow C) \rightarrow (A \rightarrow B) \rightarrow C$
17. $(A \rightarrow (A \rightarrow A) \rightarrow B) \rightarrow A \rightarrow B$
18. $(A \rightarrow (B \rightarrow B) \rightarrow C) \rightarrow A \rightarrow C$
19. $(A \rightarrow A \rightarrow B) \rightarrow C \rightarrow A \rightarrow B$
20. $(A \rightarrow B \rightarrow C \rightarrow D) \rightarrow A \rightarrow A$
21. $(A \rightarrow B) \rightarrow ((A \rightarrow B) \rightarrow A) \rightarrow A$
22. $(A \rightarrow B) \rightarrow (C \rightarrow D) \rightarrow C \rightarrow C$
23. $(A \rightarrow B) \rightarrow A \rightarrow C \rightarrow D \rightarrow D$
24. $(A \rightarrow B) \rightarrow C \rightarrow D \rightarrow E \rightarrow D$
25. $A \rightarrow ((B \rightarrow C \rightarrow B) \rightarrow B) \rightarrow B$
26. $A \rightarrow ((B \rightarrow C \rightarrow B) \rightarrow D) \rightarrow D$
27. $A \rightarrow (A \rightarrow B \rightarrow C) \rightarrow B \rightarrow C$
28. $A \rightarrow (B \rightarrow B \rightarrow B) \rightarrow B \rightarrow B$
29. $A \rightarrow (B \rightarrow B \rightarrow C) \rightarrow B \rightarrow B$
30. $A \rightarrow (B \rightarrow C) \rightarrow B \rightarrow D \rightarrow D$
31. $A \rightarrow (B \rightarrow C) \rightarrow D \rightarrow B \rightarrow C$
32. $A \rightarrow (B \rightarrow C) \rightarrow D \rightarrow E \rightarrow D$
33. $A \rightarrow B \rightarrow C \rightarrow D \rightarrow E \rightarrow E$

Depth 7 — 49 exercises

1. $((A \rightarrow A) \rightarrow A) \rightarrow A$
2. $((A \rightarrow A) \rightarrow B) \rightarrow B$
3. $(A \rightarrow B) \rightarrow C \rightarrow C$
4. $A \rightarrow A \rightarrow B \rightarrow A$
5. $A \rightarrow B \rightarrow B \rightarrow B$
6. $((A \rightarrow A \rightarrow A) \rightarrow A) \rightarrow A$
7. $((A \rightarrow A) \rightarrow B) \rightarrow C \rightarrow B$
8. $(A \rightarrow B \rightarrow A) \rightarrow A \rightarrow A$
9. $A \rightarrow (A \rightarrow A \rightarrow B) \rightarrow B$
10. $A \rightarrow (B \rightarrow B) \rightarrow B \rightarrow B$
11. $A \rightarrow B \rightarrow (B \rightarrow C) \rightarrow C$
12. $A \rightarrow B \rightarrow (C \rightarrow B) \rightarrow B$

13. $((A \rightarrow (B \rightarrow A) \rightarrow A) \rightarrow A) \rightarrow A$
14. $((A \rightarrow A) \rightarrow A) \rightarrow (A \rightarrow A) \rightarrow A$
15. $((A \rightarrow A) \rightarrow B \rightarrow C) \rightarrow B \rightarrow C$
16. $((A \rightarrow A) \rightarrow B) \rightarrow (A \rightarrow C) \rightarrow B$
17. $((A \rightarrow B \rightarrow A) \rightarrow A) \rightarrow B \rightarrow B$
18. $((A \rightarrow B) \rightarrow B) \rightarrow (A \rightarrow B) \rightarrow B$
19. $(A \rightarrow (B \rightarrow A) \rightarrow B) \rightarrow A \rightarrow A$
20. $(A \rightarrow (B \rightarrow A) \rightarrow C) \rightarrow A \rightarrow A$
21. $(A \rightarrow A \rightarrow A) \rightarrow A \rightarrow A \rightarrow A$
22. $(A \rightarrow A \rightarrow B) \rightarrow A \rightarrow C \rightarrow B$
23. $(A \rightarrow A \rightarrow B) \rightarrow C \rightarrow D \rightarrow C$
24. $(A \rightarrow B \rightarrow A) \rightarrow A \rightarrow B \rightarrow A$
25. $(A \rightarrow B \rightarrow C) \rightarrow A \rightarrow B \rightarrow A$
26. $(A \rightarrow B \rightarrow C) \rightarrow D \rightarrow A \rightarrow A$
27. $(A \rightarrow B \rightarrow C) \rightarrow D \rightarrow E \rightarrow E$
28. $(A \rightarrow B) \rightarrow ((A \rightarrow B) \rightarrow A) \rightarrow B$
29. $(A \rightarrow B) \rightarrow (A \rightarrow C) \rightarrow A \rightarrow A$
30. $(A \rightarrow B) \rightarrow (B \rightarrow C) \rightarrow A \rightarrow C$
31. $(A \rightarrow B) \rightarrow (C \rightarrow A) \rightarrow C \rightarrow A$
32. $(A \rightarrow B) \rightarrow A \rightarrow (B \rightarrow C) \rightarrow B$
33. $(A \rightarrow B) \rightarrow A \rightarrow C \rightarrow D \rightarrow A$
34. $(A \rightarrow B) \rightarrow C \rightarrow A \rightarrow D \rightarrow A$
35. $(A \rightarrow B) \rightarrow C \rightarrow D \rightarrow A \rightarrow A$
36. $(A \rightarrow B) \rightarrow C \rightarrow D \rightarrow A \rightarrow B$
37. $A \rightarrow ((B \rightarrow B) \rightarrow C) \rightarrow B \rightarrow B$
38. $A \rightarrow ((B \rightarrow C) \rightarrow D) \rightarrow C \rightarrow D$
39. $A \rightarrow (A \rightarrow (B \rightarrow A) \rightarrow C) \rightarrow C$
40. $A \rightarrow (A \rightarrow B \rightarrow C) \rightarrow B \rightarrow B$
41. $A \rightarrow (A \rightarrow B) \rightarrow C \rightarrow D \rightarrow C$
42. $A \rightarrow (B \rightarrow A \rightarrow C) \rightarrow B \rightarrow C$
43. $A \rightarrow (B \rightarrow A) \rightarrow C \rightarrow B \rightarrow A$
44. $A \rightarrow (B \rightarrow C \rightarrow D) \rightarrow B \rightarrow B$
45. $A \rightarrow (B \rightarrow C \rightarrow D) \rightarrow E \rightarrow E$
46. $A \rightarrow B \rightarrow (B \rightarrow C) \rightarrow D \rightarrow C$
47. $A \rightarrow B \rightarrow (C \rightarrow D) \rightarrow C \rightarrow B$
48. $A \rightarrow B \rightarrow C \rightarrow (C \rightarrow D) \rightarrow C$
49. $A \rightarrow B \rightarrow C \rightarrow D \rightarrow E \rightarrow B$

Depth 8 — 58 exercises

1. $A \rightarrow B \rightarrow A \rightarrow A$
2. $((A \rightarrow A) \rightarrow A) \rightarrow B \rightarrow A$
3. $((A \rightarrow B \rightarrow B) \rightarrow A) \rightarrow A$
4. $((A \rightarrow B \rightarrow B) \rightarrow C) \rightarrow C$
5. $((A \rightarrow B) \rightarrow A) \rightarrow C \rightarrow C$
6. $(A \rightarrow A) \rightarrow A \rightarrow B \rightarrow A$

7. $(A \rightarrow B) \rightarrow A \rightarrow B \rightarrow B$
8. $(A \rightarrow B) \rightarrow C \rightarrow D \rightarrow D$
9. $A \rightarrow ((B \rightarrow B) \rightarrow B) \rightarrow B$
10. $A \rightarrow ((B \rightarrow B) \rightarrow C) \rightarrow C$
11. $A \rightarrow (A \rightarrow B) \rightarrow C \rightarrow A$
12. $A \rightarrow (B \rightarrow A) \rightarrow B \rightarrow B$
13. $A \rightarrow (B \rightarrow C) \rightarrow D \rightarrow D$
14. $A \rightarrow A \rightarrow B \rightarrow C \rightarrow B$
15. $A \rightarrow B \rightarrow (A \rightarrow C) \rightarrow A$
16. $A \rightarrow B \rightarrow B \rightarrow C \rightarrow B$
17. $A \rightarrow B \rightarrow C \rightarrow C \rightarrow C$
18. $((A \rightarrow B) \rightarrow A \rightarrow B) \rightarrow C \rightarrow C$
19. $((A \rightarrow A) \rightarrow (A \rightarrow A) \rightarrow B) \rightarrow B$
20. $((A \rightarrow A) \rightarrow B \rightarrow C) \rightarrow B \rightarrow B$
21. $((A \rightarrow A) \rightarrow B) \rightarrow (A \rightarrow A) \rightarrow B$
22. $((A \rightarrow A) \rightarrow B) \rightarrow C \rightarrow D \rightarrow C$
23. $((A \rightarrow B \rightarrow A) \rightarrow A) \rightarrow C \rightarrow A$
24. $((A \rightarrow B \rightarrow A) \rightarrow C) \rightarrow A \rightarrow A$
25. $((A \rightarrow B \rightarrow A) \rightarrow C) \rightarrow D \rightarrow D$
26. $((A \rightarrow B \rightarrow B) \rightarrow C) \rightarrow D \rightarrow C$
27. $((A \rightarrow B \rightarrow C \rightarrow A) \rightarrow A) \rightarrow A$
28. $((A \rightarrow B \rightarrow C \rightarrow A) \rightarrow D) \rightarrow D$
29. $((A \rightarrow B) \rightarrow C) \rightarrow B \rightarrow A \rightarrow B$
30. $((A \rightarrow B) \rightarrow C) \rightarrow D \rightarrow E \rightarrow D$
31. $(A \rightarrow (A \rightarrow A) \rightarrow A) \rightarrow A \rightarrow A$
32. $(A \rightarrow (B \rightarrow C) \rightarrow B) \rightarrow A \rightarrow A$
33. $(A \rightarrow A \rightarrow A \rightarrow B) \rightarrow A \rightarrow B$
34. $(A \rightarrow A \rightarrow A) \rightarrow B \rightarrow A \rightarrow A$
35. $(A \rightarrow A \rightarrow B) \rightarrow C \rightarrow A \rightarrow A$
36. $(A \rightarrow B \rightarrow A) \rightarrow C \rightarrow D \rightarrow C$
37. $(A \rightarrow B \rightarrow B) \rightarrow A \rightarrow B \rightarrow B$
38. $(A \rightarrow B \rightarrow C \rightarrow B) \rightarrow A \rightarrow A$
39. $(A \rightarrow B \rightarrow C \rightarrow D) \rightarrow E \rightarrow E$
40. $(A \rightarrow B) \rightarrow (A \rightarrow B) \rightarrow A \rightarrow A$
41. $(A \rightarrow B) \rightarrow (A \rightarrow C) \rightarrow A \rightarrow C$
42. $(A \rightarrow B) \rightarrow (C \rightarrow D) \rightarrow C \rightarrow D$
43. $(A \rightarrow B) \rightarrow A \rightarrow (B \rightarrow C) \rightarrow C$
44. $(A \rightarrow B) \rightarrow A \rightarrow (C \rightarrow B) \rightarrow B$
45. $(A \rightarrow B) \rightarrow A \rightarrow C \rightarrow A \rightarrow B$
46. $(A \rightarrow B) \rightarrow C \rightarrow A \rightarrow D \rightarrow D$
47. $(A \rightarrow B) \rightarrow C \rightarrow D \rightarrow E \rightarrow C$
48. $A \rightarrow ((B \rightarrow A) \rightarrow C) \rightarrow B \rightarrow A$
49. $A \rightarrow ((B \rightarrow B \rightarrow B) \rightarrow B) \rightarrow B$
50. $A \rightarrow ((B \rightarrow B) \rightarrow C) \rightarrow D \rightarrow C$
51. $A \rightarrow (B \rightarrow A) \rightarrow C \rightarrow D \rightarrow C$
52. $A \rightarrow (B \rightarrow C \rightarrow B) \rightarrow B \rightarrow B$

53. $A \rightarrow (B \rightarrow C \rightarrow D) \rightarrow E \rightarrow A$
54. $A \rightarrow B \rightarrow (B \rightarrow B \rightarrow C) \rightarrow C$
55. $A \rightarrow B \rightarrow (C \rightarrow C) \rightarrow C \rightarrow C$
56. $A \rightarrow B \rightarrow (C \rightarrow D \rightarrow C) \rightarrow B$
57. $A \rightarrow B \rightarrow C \rightarrow (C \rightarrow D) \rightarrow D$
58. $A \rightarrow B \rightarrow C \rightarrow (D \rightarrow C) \rightarrow C$

Threshold and post-threshold starters

The exact threshold is:

Permutation P — first appears at MP-depth 9, size 11

1. $(A \rightarrow B \rightarrow C) \rightarrow B \rightarrow A \rightarrow C$

After the exhaustive size-11 below-threshold frontier, this is the point where I stop the exhaustive print and switch to starters from the same-or-harder region.

Post-threshold starter set

1. $A \rightarrow B \rightarrow A \rightarrow B$
2. $((A \rightarrow A) \rightarrow A) \rightarrow A \rightarrow A$
3. $((A \rightarrow A) \rightarrow B) \rightarrow A \rightarrow B$
4. $(A \rightarrow A \rightarrow B) \rightarrow C \rightarrow C$
5. $(A \rightarrow B \rightarrow B) \rightarrow B \rightarrow B$
6. $(A \rightarrow B) \rightarrow A \rightarrow A \rightarrow A$
7. $(A \rightarrow B) \rightarrow A \rightarrow B \rightarrow A$
8. $A \rightarrow ((B \rightarrow A) \rightarrow B) \rightarrow B$
9. $A \rightarrow ((B \rightarrow A) \rightarrow C) \rightarrow C$
10. $A \rightarrow (A \rightarrow B) \rightarrow A \rightarrow A$
11. $A \rightarrow (A \rightarrow B) \rightarrow A \rightarrow B$
12. $A \rightarrow (A \rightarrow B) \rightarrow C \rightarrow C$
13. $A \rightarrow (B \rightarrow A) \rightarrow B \rightarrow A$
14. $A \rightarrow (B \rightarrow A) \rightarrow C \rightarrow A$
15. $A \rightarrow (B \rightarrow C) \rightarrow D \rightarrow A$
16. $A \rightarrow B \rightarrow (A \rightarrow C) \rightarrow C$
17. $A \rightarrow B \rightarrow (C \rightarrow A) \rightarrow A$
18. $A \rightarrow B \rightarrow A \rightarrow C \rightarrow A$
19. $A \rightarrow B \rightarrow C \rightarrow B \rightarrow B$
20. $(A \rightarrow B) \rightarrow (A \rightarrow A) \rightarrow A \rightarrow A$
21. $(A \rightarrow B) \rightarrow (C \rightarrow A) \rightarrow C \rightarrow C$
22. $(A \rightarrow B) \rightarrow A \rightarrow (A \rightarrow C) \rightarrow A$
23. $(A \rightarrow B) \rightarrow A \rightarrow B \rightarrow C \rightarrow B$
24. $(A \rightarrow B) \rightarrow A \rightarrow C \rightarrow C \rightarrow C$
25. $(A \rightarrow B) \rightarrow C \rightarrow D \rightarrow E \rightarrow E$

Open questions and limitations

The strongest limitation is deliberate: the printed bank is **not the full raw set** of all formulas below P . The full raw set through depth 8 contains **8,531** non-axiom schemata, which is far too large to be useful as a human exercise sheet.

What *is* exhaustive here is the following:

- every non-axiom theorem schema of size ≤ 11 with minimal MP-depth < 9
- ordered exactly by minimal MP-depth, then size, then lexicographically

That cutoff is also where the search starts to become noticeably repetitive. If you want the next installment in exactly the same style, the clean continuation is:

- keep the same K, S system,
- stay with the same ordering rule,
- and print the **size-13 frontier** at depths 8 and 9 , where the next serious expansion begins.

¹ www3.cs.stonybrook.edu

<https://www3.cs.stonybrook.edu/~cse541/chapter8.pdf>